Spoken: Detection of AI-Generated Speech Using Acoustic Features and Machine Learning

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***Abstract*—This work investigates the discrimination of bona fide and synthetic speech using handcrafted acoustic features and conventional machine-learning classifiers. The Spoken system compares mel-frequency cepstral coefficients (MFCC), linear-frequency cepstral coefficients (LFCC), spectral and time-domain statistics, and an MFCC-plus-spectral representation with support vector machines (SVM) and Random Forest classifiers. The recorded evaluation uses 12,000 recordings from 2,000 matched LJSpeech source utterances and five WaveFake generators. Source-group partitioning keeps real recordings and their generated counterparts together, while grouped cross-validation selects models using training data only. LFCC with an RBF SVM obtains the highest validation F1 score of 97.86%. On the standard held-out test, the selected model achieves 97.29% accuracy, 98.38% AI-class F1, 0.9890 ROC-AUC, and 4.58% approximate equal error rate. In a separate experiment, HiFi-GAN is excluded from training and evaluated with real speech from held-out source groups. Accuracy decreases to 91.88%, F1 to 91.70%, and ROC-AUC to 0.9733, while equal error rate increases to 8.25%. The results support LFCC as an effective representation within the evaluated setting and demonstrate useful but incomplete transfer to one unseen generator. The single-speaker corpus and limited holdout coverage constrain broader robustness claims.**

***Index Terms*—Speech deepfake detection, acoustic features, LFCC, SVM, WaveFake.**

I. Introduction

Neural speech generation makes it increasingly difficult to distinguish human recordings from synthetic waveforms by listening alone. Audio deepfake detection therefore requires evaluation of cues that can persist even when generated speech is perceptually convincing. WaveFake provides a dataset and baseline framework for studying this problem through signal-processing analysis [1].

Spoken examines whether conventional acoustic representations can discriminate real and generated speech without training a neural feature extractor. Its contributions are a comparison of four feature representations with two classifier families, training-only model selection under source-group separation, and an evaluation with one generator excluded from training. The objective is a controlled course-project comparison, rather than a new state-of-the-art detector. Only the implemented MFCC, LFCC, spectral-statistics, and combined-feature experiments are reported; the project does not evaluate a spectrogram CNN or CQCC system.

II. Background and Related Work

Speech deepfakes can be produced through text-to-speech synthesis or voice conversion. This study focuses on the LJSpeech-based WaveFake generators. Cepstral representations summarize short-time spectral structure. MFCC represents the spectrum through a mel-frequency filterbank, whereas LFCC uses uniformly spaced linear-frequency filters. Their different frequency resolutions motivate an empirical comparison; neither representation is assumed to be superior in advance. Temporal derivatives and utterance-level statistics provide fixed-dimensional inputs to conventional classifiers.

Frank and Schönherr introduced WaveFake to support research on generated-audio detection [1]. This project uses five of its LJSpeech-based generator sets. The underlying LJSpeech corpus contains 13,100 recordings of one speaker [2]. Holding out original utterances prevents their real and generated versions from crossing partitions, but does not constitute an unseen-speaker test. A generator holdout addresses a different question: whether detection transfers to a synthesis method absent from training.

III. Methodology

A. Dataset and Source Grouping

The project metadata contain 78,600 distinct paths: 13,100 real recordings and 13,100 generated recordings for each of HiFi-GAN, MelGAN, Multi-Band MelGAN, Parallel WaveGAN, and WaveGlow. Each original LJSpeech utterance ID defines a group containing one real and five generated versions. Real speech has label 0 and AI-generated speech label 1.

The saved standard experiment contains 2,000 source groups and 12,000 recordings: 9,600 training recordings from 1,600 groups and 2,400 test recordings from 400 groups. These counts describe the evaluated subset, not the complete metadata index. The code supports reproducible sampling of complete groups; the original subset command and selected-ID manifest are not retained.

`GroupShuffleSplit` reserves 20% of source groups using seed 42. The code checks that training and test group sets do not overlap. Model selection uses `GroupKFold` within the training partition; the saved standard-model configuration records three folds. Thus, related versions of one utterance remain together during both final splitting and cross-validation.

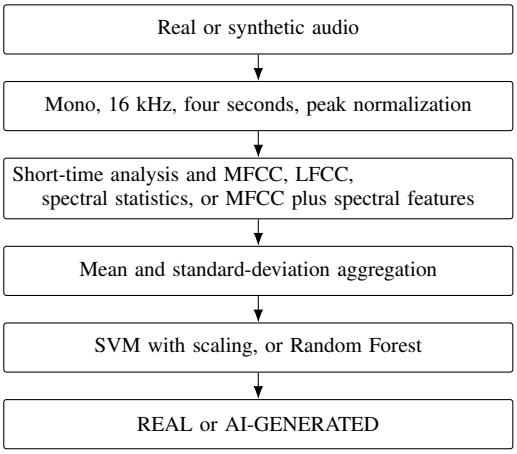


Fig. 1. Implemented processing pipeline. Each candidate uses one feature representation; selection is performed on training folds before held-out evaluation.

B. Preprocessing and Features

Fig. 1 summarizes the implemented pipeline. Audio is converted to mono and resampled to 16 kHz, then truncated to the first four seconds or zero-padded to 64,000 samples. Peak normalization is applied when the maximum absolute amplitude exceeds 10^{-9}. The code does not apply explicit silence removal, denoising, or augmentation. Short-time analysis uses a 400-sample window (25 ms), a 160-sample hop (10 ms), and a 512-point FFT. Audio processing uses librosa [3].

MFCC extraction retains 20 coefficients, their first derivatives, and their second derivatives. The LFCC implementation computes the power spectrum, applies 40 linearly spaced triangular filters over 0–8 kHz, floors energies at 10^{-12}, takes the natural logarithm, and applies an orthonormal type-II discrete cosine transform. Its first 20 coefficients and two derivative orders are retained. For both cepstral families, the mean and standard deviation of each trajectory yield $20 \times 3 \times 2 = 120$ features.

The spectral representation pools six trajectories: spectral centroid, bandwidth, 85% roll-off, zero-crossing rate, root-mean-square magnitude, and spectral flux. Flux is the Euclidean change between consecutive magnitude spectra. Two statistics per trajectory yield 12 features. The combined representation concatenates MFCC and these statistics into 132 features; it does not include LFCC. Pooling produces compact vectors but discards detailed frame ordering.

C. Classifiers and Model Selection

The implementation uses scikit-learn [4]. Both classifier families use balanced class weights. The SVM pipeline fits `StandardScaler` within each training fold and searches an RBF kernel with $C \in \{1, 10\}$ and $\gamma \in \{\texttt{scale}, 0.01\}$. Random Forest searches 200 or 400 trees, unrestricted depth or depth 20, and minimum leaf size 1 or 2, without scaling.

Eight feature–classifier candidates are compared by their best grouped cross-validation AI-class F1 score. The candidate with the highest training-validation score is selected before standard test evaluation. The selected LFCC-SVM uses $C = 10$ and $\gamma = \texttt{scale}$. Test metrics are not used for model selection. The saved bundle contains the standard model and its preprocessing configuration; the unseen-generator experiment trains a separate estimator.

IV. Experiments and Results

A. Evaluation Protocol

The standard test contains 400 real and 2,000 generated recordings. Accuracy, precision, recall, F1, ROC-AUC, and approximate equal error rate (EER) are available in the saved results. Precision, recall, and F1 refer to the positive AI class. ROC-AUC uses SVM decision margins, not calibrated probabilities. The EER routine selects the sampled ROC point minimizing $|\mathrm{FNR} - \mathrm{FPR}|$ and averages its two rates; it does not interpolate an exact crossing. Confusion matrices use classifier predictions rather than the EER threshold.

For generalization testing, the same source-group split is used, but HiFi-GAN is removed from training. The selected LFCC-SVM family is fixed and its hyperparameters retuned on 8,000 recordings from 1,600 groups: real speech and the four remaining generators. Testing retains only real and HiFi-GAN recordings from the 400 held-out groups, giving 400 recordings per class. HiFi-GAN and the test source utterances are therefore absent from this experiment's training data. The selected parameters remain $C = 10$ and $\gamma = \texttt{scale}$.

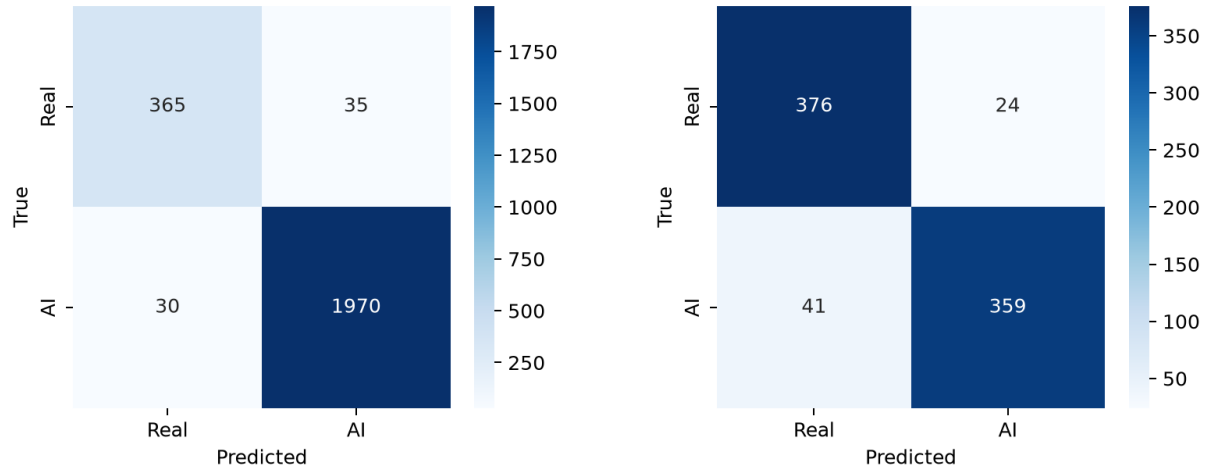
TABLE I
Training-only candidate comparison. Scores are grouped cross-validation AI-class F1, not held-out test results.

Feature	SVM F1 (%)	Random Forest F1 (%)
MFCC	88.85	90.88
LFCC	**97.86**	96.81
Spectral	74.69	90.12
MFCC + spectral	89.35	90.92

B. Candidate Comparison

Table I reproduces the model-selection scores. LFCC is strongest with both classifiers. Its SVM F1 exceeds MFCC-SVM by 9.01 percentage points and LFCC–Random Forest by 1.04 points. Random Forest performs better for the other feature families, so neither classifier dominates across all representations. Adding spectral statistics to MFCC improves SVM F1 by 0.50 points and Random Forest F1 by only 0.04 points.

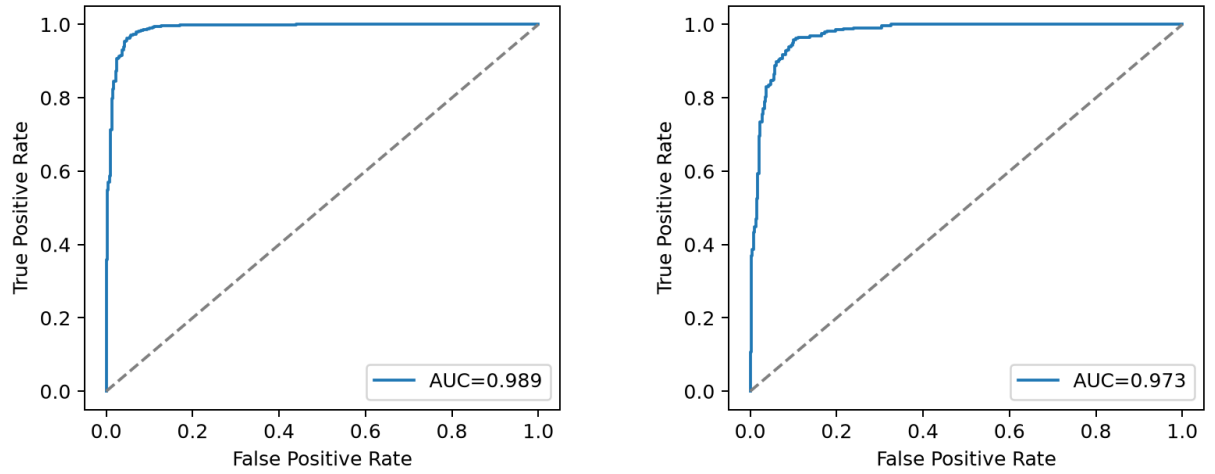
Linear-frequency resolution is a plausible explanation for LFCC's advantage, but no frequency-band ablation identifies the responsible cues. The comparison establishes performance differences between complete representations, not a causal role for a particular frequency region. Fold-level variability and repeated-seed trials are unavailable, so small differences should not be interpreted as statistically significant.



(a) Standard held-out test

(b) Unseen HiFi-GAN test

Fig. 2. Confusion matrices with true labels in rows and predicted labels in columns. Tests (a) and (b) contain 2,400 and 800 recordings, respectively, with different class mixtures.



(a) Standard held-out test

(b) Unseen HiFi-GAN test

Fig. 3. ROC curves computed from SVM decision margins; dashed lines indicate chance ranking. Legends retain the original three-decimal AUC values; Table II reports four decimals.

TABLE II
HELD-OUT RESULTS FOR THE SELECTED LFCC-SVM FAMILY. AUC IS UNITLESS; OTHER PERFORMANCE MEASURES ARE PERCENTAGES.

Measure	Standard	Unseen HiFi-GAN
Train / test recordings	9,600 / 2,400	8,000 / 800
Accuracy (%)	97.29	91.88
AI precision (%)	98.25	93.73
AI recall (%)	98.50	89.75
AI F1 (%)	98.38	91.70
ROC-AUC	0.9890	0.9733
Approximate EER (%)	4.58	8.25

C. Standard and Unseen-Generator Performance

Table II reports the saved final metrics. The standard model correctly classifies 2,335 of 2,400 recordings. Fig. 2 shows 365 true negatives, 35 false positives, 30 false negatives, and 1,970 true positives. The real-speech false-alarm rate is 8.75%, whereas the AI miss rate is 1.50%. Because the test has five generated recordings per real recording, AI-class F1 alone gives an incomplete account of class performance. Balanced accuracy derived from the matrix is 94.88%; an always-AI baseline would obtain 83.33% ordinary accuracy.

The unseen-generator model correctly classifies 735 of 800 recordings, with 376 true negatives, 24 false positives, 41 false negatives, and 359 true positives. Its real false-alarm rate is 6.00% and HiFi-GAN miss rate is 10.25%. Relative to standard evaluation, AI recall decreases by 8.75 percentage points and EER rises from approximately 4.58% to 8.25%. The retained AUC of 0.9733 suggests useful separation beyond the training generators, while the increased miss rate exposes

a generalization gap.

Fig. 3 shows the tradeoff between true-positive and false-positive rates as the decision threshold varies. High AUC can coexist with missed synthetic speech at the classifier threshold; ranking quality and fixed-threshold decisions describe different aspects of performance. The approximate EER provides an additional threshold-based summary, not a calibrated deployment threshold.

The feature–classifier family was chosen in the standard comparison, which includes HiFi-GAN. The holdout therefore evaluates transfer of the retrained selected family, not fully generator-blind architecture selection. This comparison also changes training composition, test generators, and test class balance. The accuracy difference is therefore not a generator-only effect measured on an identical test population. Testing both trained systems on the same real/HiFi-GAN subset and repeating the holdout for each generator would strengthen the assessment.

D. Limitations

The evidence covers a single-speaker corpus, one evaluated subset, and one generator holdout. Unseen speakers, other languages, noise, compression, and recording-channel variation are not tested. Per-record predictions and split manifests are not retained, limiting detailed error analysis and exact reproduction. The supplied archive contains the metadata and saved results but not the original experiment audio. Artificial smoke-test signals are excluded from all reported scientific metrics.

V. Team Contribution

Mohammad Sayyad contributed to dataset preparation and organization, source-group construction and data splitting, preprocessing, and the implementation and evaluation of MFCC and spectral feature extraction.

Abdallah Hasan contributed to LFCC feature extraction, implementation and tuning of the SVM and Random Forest classifiers, candidate-model comparison, and standard held-out evaluation, including performance metrics and result visualization.

Mahmoud Suliman contributed to overall system integration and experimental workflow, implementation and verification of the unseen-generator HiFi-GAN experiment, final model-selection verification, analysis of confusion matrices, ROC and error metrics, debugging and validation of the complete pipeline, and coordination of the final results and documentation.

VI. Conclusion

Spoken compares four handcrafted acoustic representations with SVM and Random Forest under source-group-aware evaluation. LFCC-SVM is the strongest candidate and reaches 97.29% standard-test accuracy and 98.38% AI-class F1. Excluding HiFi-GAN from training retains 91.88% accuracy but increases missed generated speech, supporting limited transfer rather than universal deepfake detection. Future work should extend generator holdouts and speaker diversity, evaluate channel distortions, and preserve per-record scores for reproducible error analysis.

AI Assistance Disclosure

OpenAI ChatGPT and Codex [5], [6] supported code review, debugging, artifact verification, and report preparation. Reported experimental results were taken from the existing project outputs.

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